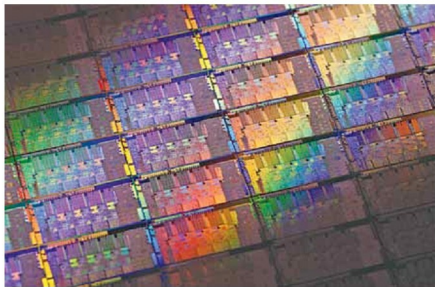


Sandy, Ivy and Haswell: Tock, Tick, Tock

A rundown of Intel's CPU microarchitectures to clear the mist surrounding them



Sandy Bridge wafer by Intel

DILIN ANAND

Things tend to get confusing when a chip manufacturer releases a new version just as a device powered by the previous release reaches your hands. Intel's Core-i branding gets no clearer with each subsequent release of codenamed chips. Since Intel has just launched the die-shrink of the current Sandy Bridge microarchitecture, we realised it is high time we did a rundown of Intel's CPU microarchitectures and cleared the mist surrounding them.

What are i3, i5 and i7?

These three modifiers form the different line-ups of Intel's processors.

Core i3. This line-up targets the low-end market. These processors have a lower average thermal design point (TDP) rating which makes them

perform cooler, albeit with lower performance. The TDP is the amount of power that the cooling system in a computer should dissipate to ensure proper functioning of the system. The i3 processors are usually restricted to two cores.

Core i5. This is the mainstream variant of Intel's processors. These processors have a higher average TDP and a larger L3 cache. The cache is an auxiliary memory from which high-speed retrieval of data is possible. This is possible due to its placement on the processor die rather than external to it. Computers have different levels of cache memory: L1, L2 and L3. As the levels increase, the time to retrieve data from them increases along

with the amount of data they can store. The i5 processors are available with both two and four cores and are hyper-threading enabled.

Core i7. This is a high-end offering from Intel. Core i7 variants may feature two, four or even six cores along with a larger L3 cache.

Sandy Bridge

Sandy Bridge is the codename for the current serving of microprocessors from Intel. It is based on a 32nm manufacturing process. Intel claims that it offers up to 14 per cent increased performance clock-for-clock over the previous Nehalem microprocessors.

The Sandy Bridge naming scheme continues with the same Core-i branding as before. But one minor difference helps us to differentiate the generations. As you can see from Figs 1 and 2, the generation of a processor can be found out by checking the digit in the thousand's place that comes after the modifier. So an Intel Core i7-2600K is a second-generation processor, while an Intel Core i7-620M is a first-generation processor (no digit means first generation).

The suffix 'K' signifies that the pro-

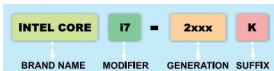


Fig. 1: Intel's second-generation processor naming

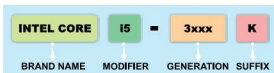


Fig. 2: Intel's third-generation processor naming